

Counting Beats the Organ

A Negative Result on Retrieval-Augmented Causal Banks,
and the Baseline We Never Ran

Built by Claude instances, steered by a human, measured by machines
Systems: DECEPTICONS (kernel) · CHRONOHORN (runtime) · HEINRICH (forensics)

Every number below was co-witnessed by two independent instruments
and none of them was worth having.

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Abstract

We set out to buy bits. A frozen-substrate causal bank—a recurrent body whose kernel is fixed at initialization and whose readout is trained—was to be given a memory organ: a k -nearest-neighbour datastore over its own hidden states, mixed into its output distribution. Over five store sizes from 8M to 150M bytes we measured the organ’s value, co-witnessing every rung to the fourth decimal across two independent implementations, and it behaved: the lift grew monotonically, the optimal mixing weight rose with store size, an oracle bound priced the ceiling, and a four-corner control experiment cleanly separated what retrieval was buying (calibration repair) from what it was not (cross-domain knowledge). Then we ran, at the operator’s suggestion, the control we should have run first. **An order-7 byte n -gram—a count table, ninety seconds of CPU—dominates the organ at every store size, does not saturate, and, alone, out-predicts the trained model itself** (2.0197 vs. 2.0367 bpb). Our best mix, 150M keys and 38 GB of RAM at 1.9530 bpb, is beaten by a count table holding a third of the bytes (1.8681 bpb). Independently and from the opposite end, a weight-level measurement finds that the trained readout places *less* energy in the directions that decode the lagged past than a random matrix would (0.40–0.43× chance): training does not merely fail to use long context—it turns away from it. The two results are the same fact seen from two sides, and the fact is that the body was not using its long context, so an organ that supplies more of it could not have helped. We report the negative result, the one claim that survives it (a byte unigram—256 numbers—captures 93.6% of what a 64-million-key archive buys on foreign text), and the methodological failure that produced a month of honest, precise, irrelevant measurement: every pre-registration we wrote carried a literature line and a kill line, and none carried a *baseline* line.

1 What This Document Is

This is a negative result, reported in full, by the two agents who produced it and the human who ended it with one question.

Three repositories are involved and it matters which does what. DECEPTICONS supplies the kernel: the causal-bank model, its frozen linear substrate, and the retrieval primitives. CHRONOHORN supplies the runtime: training, evaluation harnesses, and the datastore experiments. HEINRICH is the forensic instrument: it measures what a model *computes*—residual geometry, state information, weight structure—rather than what it outputs, and it was used here both to co-witness CHRONOHORN’s claims on an independent implementation and to make measurements no loss curve can make.

The campaign had a target: 1.0 bits per byte on **enwik8**-family text with a small byte-level model. It had a strategy: since the substrate is frozen and cannot learn to store the past, *give*

it the past, as an external archive of its own states. It had a protocol we were proud of—pre-registration before every run, a declared kill line, two independent witnesses agreeing to the fourth decimal before any number entered the record. Section 3 reports what that protocol produced. Sections 4 and 5 report the two measurements that destroyed it. Section 8 is the part we think is actually worth publishing.

2 The Bet

The causal bank is a recurrent architecture whose state is a bank of exponential moving averages of a projected byte embedding, with decay rates fixed at initialization across a spectrum of half-lives. The substrate is *frozen*: it is never trained. Only a readout—an MLP reading the concatenation of the state and a local embedding—learns. The architectural bet is that a rich enough fixed temporal basis, plus a trained readout, can substitute for a learned recurrence, and that this buys enormous efficiency.

The bet’s obvious weakness is memory. A frozen EMA bank cannot choose what to keep. Long-range structure that a transformer holds verbatim in its KV cache, or that an LSTM learns to gate, must here be recoverable from a fixed linear mixture of the past—or not at all. This is the gap the retrieval organ was meant to close: if the body cannot hold the past, let it *look the past up*.

Concretely: run the body over a large corpus, store each state vector alongside the byte that followed it, and at inference retrieve the k nearest stored states to the current one, vote over their successor bytes, and mix that vote into the body’s output distribution with weight λ . This is k NN-LM, transplanted from transformers onto a causal bank, and the ladder that follows measures what it buys.

3 The Campaign, As It Looked From Inside

3.1 The ladder

The base model is `fo-noadapt-enwik90-50k`: a byte-level causal bank trained on the first 90M bytes of `enwik9`, evaluated on held-out `enwik8`-family windows at **2.0367 bpb**. Every rung below stores states from the same corpus and queries held-out windows; every rung was run twice, once in CHRONOHORN’s harness and once in HEINRICH’s independent implementation with its own cross-entropy accounting, and the two agreed to the fourth decimal at all five rungs.

store	Δ vs. base	λ	mix
8M	−0.0144	0.05	2.0223
16M	−0.0187	0.05	2.0180
32M	−0.0367	0.10	2.0000
64M	−0.0488	0.10	1.9880
94M	−0.0686	0.20	1.9682
150M	−0.0837	0.50	1.9530

Table 1: The kNN ladder. Monotone, double-witnessed, and beside the point.

goldens: docs/data/knn-ladder-*.txt base guard: 2.0367 exact or abort

It behaved exactly as a working mechanism should. The lift grew with store size. The optimal mixing weight λ rose with it—roughly doubling per two store doublings—which we read as a demand curve: the bigger the archive, the more the model wants to listen to it. Two

independently constructed calibration grids, with different geometries, never disagreed about the shape of that curve.

3.2 The oracle purse, and a case for a port

For each rung we computed a per-position oracle over the two endpoints {base, pure k NN vote}: an upper bound on what *any* gating function $\lambda(x)$ could collect. The purse grew with the store (0.2610 bpb at 8M, 0.3788 at 94M, 0.4377 at 150M) while the scalar mixture’s take of it plateaued around 19%. We called this the *widening gap* and it became the strongest quantitative argument for the next experiment—an attention port that would read the retrieved neighbours directly instead of averaging them into a vote. The argument had the pleasing property of not depending on the oracle being achievable: it rested only on the mixture demonstrably failing to track a growing purse.

It was an artifact. See §4.

3.3 The relevance square, and the surface table

The one part of the campaign that still stands as *method* is the decomposition. A raw retrieval delta conflates two different purchases: an archive of *any* text repairs a badly calibrated base (*smoothing*), and an archive of *relevant* text supplies content the base lacks (*knowledge*). Running a store of irrelevant text as a control isolates the first, and the difference gives the second.

Across four held-out domains, with an `enwik8` store as the irrelevant control:

fold	base	total	smoothing	knowledge
code	4.9036	−2.9080	−1.1837	− 1.7243
math	−	−0.7666	−0.3233	− 0.4433
wikiml	−	−0.4566	−0.5740	+0.1174
world	4.8797	−0.4204	−0.6245	+0.2041

Table 2: The surface table. On foreign prose, *relevance subtracts*: the irrelevant store beats the relevant one. On templated text (code, math) the knowledge column is real and large.

```
goldens: docs/data/code-fold-cowitness.txt, knn-square-*.txt    code fold + control
co-witnessed at the 4th decimal
```

The sign of the knowledge column separates two regimes cleanly, and a cheap label-free discriminator—is the archive alone a better predictor than the base? —predicted that sign correctly in all four folds without needing the control at all. The policy sentence we drew from it was: *build stores for templated domains; calibrate for foreign ones*.

Hold onto the word *calibrate*. It is about to become the whole story.

4 Refutation I: A Byte Count Table Dominates the Organ

The operator asked why we had never used n -grams.

The question stings because the answer is not “we lacked them.” DECEPTICONS ships `OnlineCausalMemory`, a causal n -gram accumulator. The polyhash module ships `NgramTable`. We had count tables in both repositories the entire time and never pointed one at the organ.

So we built the control properly: an interpolated byte n -gram over orders $0..K$ with Witten-Bell-style backoff and 64-bit rolling context hashes, trained on *the same store bytes* a k NN rung uses, scored on *the same query windows*, mixed into *the same base logits*. Two numbers decide it: which is the better standalone predictor, and which buys more when mixed.

4.1 Standalone

predictor	bpb
best byte n -gram, 8M store (order 7, $\alpha=64$)	2.3128
state- k NN organ alone, 8M store	2.9743
state- k NN organ alone, 150M store	2.3892
the trained base itself	2.0367
n -gram alone, 94M store	2.0197

Table 3: An order-7 count table on 8M bytes beats the retrieval organ on 150M bytes. At matched store size it is not close. And at 94M, the count table out-predicts *the trained model*.

chronohorn: python/chronohorn/eval/ngram_control.py ruling b2815ef5

4.2 Mixed—the decisive test

Standalone strength is not the claim the ladder made. The ladder measured what the organ buys *mixed into a trained base*, and complementarity is not strength: the base is a neural LM trained on this very corpus and may already contain every n -gram statistic there is, in which case mixing them back in would buy nothing. That was the organ’s last defence, and it fails.

store	organ Δ	n -gram Δ	ratio
8M	−0.0144	− 0.0515	3.6×
94M	−0.0686	− 0.1686	2.5×

Table 4: Same body, same store bytes, same canonical eval windows (the base must reproduce 2.0367 exactly or the run aborts). The n -gram does not saturate: it scales alongside the organ from a far higher base, so the organ never catches up.

chronohorn commit bd68f85 ruling abff0759 (reset)

base + n -gram at 94M = 1.8681 bpb beats **base + organ at 150M = 1.9530 bpb**—our best-ever mix, beaten by a count table with a third of the bytes, in ninety seconds of CPU and 5.3 GB, against hours of GPU and 38 GB.

4.3 What this retracts

The *widening gap*—our strongest argument for the attention port—dies here, and it dies in the most embarrassing way available: the scalar mixture collects the n -gram at **41.4%** of its purse where it collects the organ at **18.1%**. The mixture was never structurally blind. It was being handed a weak, badly calibrated predictor. The “unrealized purse” we were pointing at was *the organ’s own incompetence*, wearing the costume of a mechanism.

Dead, therefore: the five-rung ladder as a value claim; the product-quantization tier that would have built 900M keys; the attention port’s premise as written. The ladder measured the *cost* of retrieval, not its value.

4.4 The near-miss that should frighten us most

The first pass of this control used $\alpha = 1.0$, untuned, and reported that *the organ beats the n -gram by 0.345 bpb*. Pushing α until the curve turned inverted the verdict by **1.03 bpb**.

We came within one lazy hyperparameter of publishing the opposite conclusion, with four decimals and two witnesses behind it. **An untuned control is not a control; it is a prop.** The rule we extract in §8 is stronger than “run a baseline,” and this is why.

5 Refutation II: The Readout Evacuates the Past

The second refutation was not looking for the first. It came from HEINRICH, from a weight measurement, and it arrived on the same day.

5.1 The question, and the null

The frozen substrate holds the past in a fixed linear mixture. *Which directions of the state actually decode the lagged past?* We answered it by frequency-weighted class-mean SVD on a sequence-mode capture of the body’s own frozen initialization—for each lag ℓ , the directions along which the state separates by the byte that occurred ℓ steps ago. For the experimental body (512 modes) this yields a rank-7 subspace: 5 directions at lag 64, 2 at lag 512, merged and orthonormalized.

Then, the crucial part, and the part we nearly got wrong. *How much energy would a readout put in those 7 directions by accident?* An isotropic (Xavier) init already places a fraction r/D of its weight energy in *any* fixed rank- r subspace. Here $7/512 = \mathbf{0.0137}$. The measured value on the real untrained body is 0.0131. **An untrained readout is already as “aligned” with the retrospective subspace as chance allows**, and any alignment claim not quoted against this floor is unreadable. (We initially carried a floor of 0.109 from a 64-mode toy—a number correct in its own context and meaningless in this one. The error is the same class as everything else in this paper.)

5.2 The measurement

Four training arms, identical but for how the readout’s state-facing block is seeded: naive (Xavier), zeroed, and two *projected* arms that rotate a prescribed fraction of the block’s energy into the retrospective subspace while conserving its Frobenius norm exactly— $1/3$ (a $24.4\times$ chance concentration) and 0.05 ($3.7\times$).

checkpoint	energy in subspace	$\times$ chance	$\ \text{block}\ $	eff. rank
init (pre-hook)	0.0131	0.96	17.73	170.60
A naive @1k	0.0087	0.64	31.07	40.27
A naive @5k	0.0056	0.41	48.36	33.44
B zeroed @1k	0.0104	0.76	23.50	18.91
B zeroed @5k	0.0059	0.43	41.15	25.94
C reserve @1k	0.0727	5.32	31.03	40.33
C reserve @5k	0.0138	1.01	48.50	36.04
C2 mid @1k	0.0119	0.87	31.03	40.13
C2 mid @5k	0.0054	0.40	47.35	35.50

Table 5: The readout’s state-facing block, projected into the rank-7 retrospective subspace. Chance is $1.00\times$.

golden: docs/data/e4-init-geometry-survival.txt instrument: heinrich
profile-cb-subspace-energy ruling 75906b12

Two things are visible and only one of them was predicted.

The imposed geometry dissolves. Arm C begins at $24.4\times$ chance and decays to $5.32\times$ by step 1000 and $1.01\times$ by step 5000. Whatever we seed, training returns the readout to chance. The final losses of all four arms (2.4107, 2.4103, 2.4061, 2.4107) lie inside the ~ 0.004 bpbinit-noise floor: *the init geometry does not survive training*. That much we expected to read off the loss curves; here it is read off the weights.

But the trained readout does not stop at chance—it goes below it. Arms A, B and C2 all finish at $0.40\text{--}0.43\times$ chance. *The trained readout places less energy in the directions that decode the lagged past than a random matrix would.* It does not merely fail to use long context. It *turns away from it*, systematically, and it spends its training steps doing so.

This is the same arrow the forensic program had already found in the state’s information content—that training moves the body from an archive with its head in the past toward attention on the present edge—now visible in a second, independent surface: the readout’s own weights.

5.3 What we withdrew

We had a story for why the aggressive projected arm (C) was slowest early: over-concentration, i.e. rank collapse in the readout. It is *not supported*. At step 1000 arm C’s effective rank (40.33) equals naive’s (40.27). And our evidence cannot see the regime where the effect lived—C’s early penalty is at steps 50–200, the earliest checkpoint is 1000, and by then the loss curves have converged. The mechanism of C’s early penalty is **unknown**. We record it as unknown, having proposed it loudly and having had a collaborator build an experimental arm around it.

6 Convergence: The Body Is Not Using Its Long Context

Put the two refutations side by side.

- A 7-byte count table out-predicts the entire trained body (2.0197 vs. 2.0367).
- The trained readout actively turns away from long-lag directions ($0.41\times$ chance).

They are the same fact from opposite ends. The first says: whatever the body earns above the count table’s reach, it does not earn from context longer than about seven bytes. The second says: the body’s readout has *decided* not to look there. The frozen substrate’s long-timescale modes—the entire architectural premise, the thing the retrieval organ was built to supplement—may be dead weight the readout has learned to ignore.

This reframes the negative result. The organ did not fail because retrieval is worthless, or because k NN is the wrong retrieval mechanism. **It failed because it was supplying more of a resource the body had already refused.** No better organ, no attention port, no product-quantized 900M-key store would have changed that, and each was on our roadmap.

7 What Survives

7.1 The unigram prices the smoothing column

The single most useful number the campaign produced is the one that embarrassed it. On a foreign fold (enwik8 store, world queries, base 4.8797), we scored the archive against the store corpus’s **byte marginal**—256 numbers, no keys, no search—on identical positions, identical base logits, with λ tuned on the same calibration split:

mixed with	Δ	mix
the archive (64M keys)	−0.6245	4.2553
a byte unigram (256 numbers)	−0.5843	4.2955

Table 6: The unigram captures **93.6%** of what a 64-million-key archive buys. Co-witnessed on two instruments; the figure reproduces to the digit.

golden: docs/data/marginal-unigram-cowitness.txt ruling ac7bc509

This does not kill the smoothing/knowledge decomposition—it *prices* it. What we had been calling “calibration repair” was never retrieval at all: it is an overconfident base being pulled toward byte frequencies, and *anything* that pulls it there works. An archive is an extraordinarily expensive way to compute a unigram. The policy sentence survives with a cost column attached: for a smoothing fold, do not build a store—ship 256 numbers.

7.2 The decomposition, as method

The square and the smoothing/knowledge split are claims about *mixing an archive into a base*, not about state- k NN. They survive the reset as a method, and every number in them must be re-measured with a counting archive. The open question it leaves is sharp and cheap: templated text is exactly what n -grams eat, so *does code’s −1.7243 knowledge column survive against a count-table archive?* If it does, relevance is real. If it does not, the square was about count statistics all along.

7.3 The instruments

The forensic tools outlive the hypothesis they were built to test: **profile-cb-subspace-energy** (does an init geometry survive training?), **profile-cb-retro-subspace** (which state directions decode the lagged past?), and a k NN-lift harness that now *always* emits its own unigram control. That last change is the paper’s one line of code worth copying: the refutation is no longer optional, so it cannot be forgotten.

8 Anatomy of an Error

This is the part we believe is worth publishing.

The campaign was not sloppy. It was, by every local measure, unusually careful: pre-registration before each run, a declared kill line, two independent implementations agreeing to the fourth decimal before anything entered the record, a written protocol for hardware contention, and an explicit rule that no analysis survives as a script—it becomes an instrument or it does not exist. We caught, in flight, a corpus loader silently interleaving nulls (by hashing bytes against their producing reader), a GPU throttled $8\times$ while reporting 100% utilization, and an evaluation harness in which *raising the calibration count silently drew a disjoint test set*—caught only because a guard tied the new harness to a known base value and refused to run when it drifted by 0.03 bpb.

And all of that precision was spent measuring a mechanism that a ninety-second count table beats.

8.1 The rule

No experiment without its cheapest competent baseline, tuned to its own optimum—and the baseline is tuned by whoever stands to lose from it.

Every pre-registration we wrote had a literature line and a kill line. None had a *baseline* line. The kill line asks “what result would make me abandon this hypothesis?”—and we answered it honestly every time, which is exactly why it felt like rigour. But a kill line polices the hypothesis *against itself*. It never asks the other question: *what cheap thing might already do this, and would I notice if it did?*

The second clause is not rhetoric. The untuned $\alpha=1.0$ pass had the n -gram losing by 0.345 bpb; tuning it flipped the verdict by 1.03. A baseline tuned by its proponent’s opponent is the only kind you can trust, and in a two-agent system that is a role you can actually assign.

8.2 The error class, five times in one day

Once named, the same failure was everywhere—always a number that meant one thing in one regime, quoted as if it meant the same in another:

1. An untuned n -gram ($\alpha=1.0$) reported as “the organ wins.”
2. A chance floor measured on a 64-mode toy (0.109) carried into a 512-mode body, where the true floor is 0.0137—an $8\times$ error in the null of the central alignment claim.
3. A purse-anatomy figure remembered from a letter and compared against a fresh measurement, nearly producing a spurious “whitespace collapse.”
4. Gate rules posting an 84–86% purse share on a lopsided fold—which we nearly cashed as evidence of learned routing, until we read the usage column: the gates were firing 94–96% of the time. They had degenerated to *always-retrieve*, which is near-optimal when retrieval alone beats the base. No routing was being learned at all. *Read the usage before crediting the share.*
5. An API argument written from memory of its own signature rather than copied, silently mis-passed.

8.3 The fourth decimal is not rigour

It is the *last* step of rigour, and we performed it first and hardest because it is the one that feels like virtue. Two witnesses agreeing to four decimals is a strong guarantee that a number is *correct*. It is no guarantee whatsoever that the number is *about anything*. We had a protocol optimized end-to-end for precision and containing no step whose job was relevance.

Rigour is asking what would make the entire enterprise pointless, and then spending ninety seconds of CPU to find out.

9 Threats to Validity, and What We Did Not Show

We are reporting a negative result and the temptation to over-generalize it is exactly symmetrical to the temptation that produced the positive one. So, plainly:

- **One architecture, one scale, one corpus family.** All of this is a byte-level causal bank with a frozen substrate, 0.4M–11M parameters, on **enwik**-family text. A count table beating *this* body says nothing about count tables beating a transformer, or a larger bank. Everything above ~ 1.8 bpb on this corpus is commodity statistics—PPM’s well—and our best mix (1.9530) never left the well. A model that had left it might use its long context.
- **The organ was *our* organ.** Whitened state k NN with a softmax vote is one retrieval design. A different key construction might not be dominated. We showed that *this* organ is dominated, and that the reason is upstream of the organ—but the second half of that claim rests on the readout measurement, which is a single body.

- **E4 is one seed per arm.** The early-training inversions are 46–69× the noise floor, so they should hold, but they are not replicated. The steps 50–200 regime, where the only unexplained effect lives, was never captured.
- **The context-length decomposition was never run.** Our central claim —the body is not using its long context—rests on two convergent measurements, not on a direct one. The direct measurement is stated below and it is cheap. We did not run it, and a reader is entitled to hold the claim at exactly the confidence two indirect witnesses earn.

10 Conclusion, and the One Question Left

We built a retrieval organ for a frozen-substrate causal bank, measured it across six store sizes with two independent instruments to the fourth decimal, priced its ceiling with an oracle, decomposed its value into smoothing and knowledge with a four-corner control, and designed an attention port to collect the residual. A byte count table beats it at every size, does not saturate, and out-predicts the model itself. The model, independently, turns out to place less weight on the directions that decode its own distant past than a random matrix would.

The organ was supplying more of something the body had already refused.

The question that follows is the only one we would now spend a GPU-hour on, and it was never asked because it does not presuppose the answer we wanted:

Where, if anywhere, does the body beat counting?

Decompose base-versus- n -gram performance by context length. If the body beats a count table *nowhere* beyond about seven bytes, then the frozen-substrate bet is refuted on its own terms, no store of any kind deserves another hour, and the only remaining source of non-commodity bits is the core architecture. If there *is* a horizon where the body pulls ahead, then that horizon is the entire value of the architecture, and every future bit comes from widening it.

Either answer is worth more than the ladder was. Both are cheap. Neither requires an archive.

Reproduction

Every number in this paper has a committed artifact.

claim	artifact
the ladder	heinrich docs/data/knn-ladder-*.txt
the square, surface table	heinrich docs/data/knn-square-*.txt
code fold + control (2 witnesses)	heinrich docs/data/code-fold-cowitness.txt
the unigram control (93.6%)	heinrich docs/data/marginal-unigram-cowitness.txt
retro subspace + chance floor	heinrich docs/data/e4-retro-subspace-init.txt
readout evacuation	heinrich docs/data/e4-init-geometry-survival.txt
n -gram standalone	chronohorn eval/ngram_control.py (ruling b2815ef5)
n -gram mixed (the reset)	chronohorn commit bd68f85 (ruling abff0759)
init hooks, E4 arms	decepticons commit 0ba25a0

Coda

The operator asked one question, in eight words, and it cost the campaign its premise and bought it its only durable finding. We had the tooling to answer that question on day one, in both repositories, and we spent the month racing each other to the fourth decimal instead.

The instruments were good. The witnesses were honest. The protocol was followed. And none of it was pointed at the one thing that could have shown us we were wrong, because everything in the protocol was designed to check whether we were *precise*, and nothing in it was designed to check whether we were *necessary*.

That is the result. It is negative, and it is a result.